# HASSAN AHMAD

AI / ML Engineer | Python · Computer Vision · Chatbot Development · API Integration

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## PROFILE

Software Engineering graduate with hands-on experience building machine learning, computer vision, and chatbot systems through internships at ITSOLERA, Arch Technologies, and Cognifyz Technologies. Skilled in Python, classical ML, deep learning, time-series forecasting, and REST API integration. Authored a research paper, "Deep Learning-Based Multimodal Driver Alertness Classification with Missing Modality Robustness," currently ready for IEEE submission. Passionate about developing innovative, ethical AI solutions and delivering real-world impact.

## EDUCATION

**B.Sc. Software Engineering, CGPA 3.48 / 4.0** — *Sep 2022 – Jun 2026*

*University of Malakand, Pakistan*

- Field of study: Information and Communication Technologies (EQF Level 6); coursework in machine learning, deep learning, computer vision, data structures, algorithms, and software design
- Final Year Research: "Deep Learning-Based Multimodal Driver Alertness Classification With Missing Modality Robustness" — manuscript ready for IEEE submission

## EXPERIENCE

**Machine Learning Engineer (Intern)** — *Oct 2025 – Present*

*ITSOLERA Pvt. Ltd. · Islamabad, Pakistan*

- Built an Image Classification System using traditional ML techniques: extracted HOG features, color histograms, and edge descriptors; trained SVM, KNN, and Random Forest classifiers; evaluated with accuracy, precision, recall, and confusion matrix analysis
- Developed a Time-Series Demand Forecasting System with external variables (promotions, holidays, weather, economic indicators); compared ARIMA, SARIMA, Facebook Prophet, Random Forest, and XGBoost using MAE, RMSE, and MAPE on a rolling-window validation
- Engineered an Adaptive Synthetic Data Augmentation Toolkit: implemented a closed-loop pipeline that performs error analysis on imbalanced cohorts, generates targeted synthetic samples via GANs / VAEs / SDV, and retrains the model iteratively to improve minority-class metrics
- Built a content-based Movie Recommendation System on the TMDB 5000 Movies dataset using Python, Pandas, and scikit-learn.
- Used Python, scikit-learn, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Statsmodels, XGBoost, and Matplotlib; managed all code through Git and GitHub

**Machine Learning Engineer (Intern, Remote)** — *Aug 2025 – Sep 2025*

*Arch Technologies · Pakistan*

- Built four end-to-end ML projects: House Price Prediction (regression), Email Spam Classification (NLP text classification), Iris Flower Classification (multi-class), and MNIST Handwritten Digit Recognition (computer vision)
- Handled the full pipeline for each project: data preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation using scikit-learn and TensorFlow

**Machine Learning Engineer (Intern, Remote)** *Jan 2025 – Mar 2025*
*Cognifyz Technologies*

- Delivered a four-task restaurant analytics project: Restaurant Rating Prediction, Restaurant Recommendation System, Cuisine Classification, and Location-Based Analysis
- Applied regression, classification, recommendation, and geospatial techniques in Python using Pandas, scikit-learn, and Matplotlib; documented findings and model insights for review

## S K I L L S

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| **AI / ML** | scikit-learn, TensorFlow, PyTorch, XGBoost, Statsmodels, Facebook Prophet | **Computer Vision** | OpenCV, HOG, CNNs, Image Classification, Feature Extraction |
| **Backend & API** | Python, FastAPI, REST APIs, Chatbot Development, JSON | **Data** | Pandas, NumPy, SQL, Time-Series Analysis, Feature Engineering |
| **Generative AI** | GANs, VAEs, SDV (Synthetic Data Vault), HuggingFace | **Tools & DevOps** | Git, GitHub, Jupyter Notebook, VS Code, Docker (basic) |
| **Programming** | Python, C++, SQL, Algorithms, Data Structures | **Languages** | English (C1), Urdu (Native), Pashto (Native) |

## P U B L I C A T I O N S

*"Deep Learning-Based Multimodal Driver Alertness Classification With Missing Modality Robustness."* Manuscript ready for submission to IEEE, 2026. On going

*"Intelligent Heart Murmur Detection from Phonocardiogram Signals Using Continuous Wavelet Scalograms and a Spatial-Attention Convolutional Neural Network."* University of Malakand — Software Engineering Research Paper, 2026. ongoing

## K E Y P R O J E C T S

**Multimodal Driver Alertness Classification (Research)** *2025 – 2026*
*University of Malakand — Research Project*

- Designed a deep learning system that classifies driver alertness from multiple input modalities (e.g., facial features, eye state, head pose) and remains robust when one or more modalities are missing at inference time
- Built the full pipeline in Python with PyTorch / TensorFlow and OpenCV: data preprocessing, multimodal feature fusion, missing-modality handling strategy, model training, and evaluation
- Authored full research manuscript with introduction, methodology, results, and discussion sections — currently ready for IEEE submission

**Intelligent Heart Murmur Detection (Research)** *2025 – 2026*
*University of Malakand — Research Project*

- Co-developed an automated, non-invasive heart murmur detection system that converts 1D phonocardiogram (PCG) recordings into 2D Continuous Wavelet Transform (Morlet) scalograms and classifies them as murmur present or no murmur
- Built a custom deep CNN with dual spatial-attention modules in Python (TensorFlow / Keras), benchmarked against nine transfer-learning baselines including Xception, ResNet-152, InceptionV3, MobileNetV2, and EfficientNet
- Achieved 99.53% accuracy, 100% precision, 97.56% recall, and 98.77% F1-score on the CirCor DigiScope (PhysioNet Challenge 2022) dataset; co-authored the full IEEE-style manuscript

**Generative AI Portfolio** May *2026 – June 2026*

*DecodeLabs — Generative AI Internship*

- Built a five-task generative-AI portfolio spanning system-prompt engineering, advanced image-generation prompting, retrieval-augmented generation, a multimodal video pipeline, and an AI safety & bias audit.
- Engineered a citation-forced RAG system for document question-answering that grounds every answer in its sources to minimize hallucination.
- Developed a multimodal content engine that transcribes video with Whisper and uses an LLM to auto-generate short-form video reels with captions.
- Conducted responsible-AI red-teaming and bias testing, producing guardrail templates and an audit report.

**Food Hub — Food-Delivery Website & Ordering Chatbot** Mar 2026 – Apr 2026

*Full-Stack Web Project*

- Built a full-stack food-delivery web app with a rule-based ordering chatbot (Node.js / Express backend, vanilla HTML/CSS/JS frontend).
- Implemented a 32-item menu, conversational and one-click ordering, order tracking and history, and an offline fallback that keeps the chatbot and orders working without the server.

**Nursing College Website & Chatbot** May 2026- June 2026

*Front-End Web Project*

- Developed a responsive multi-page college website with an in-browser knowledge-base assistant ("Ask Nightingale") that answers admissions, programs, tuition, and campus questions with no backend or API key.
- Added a category-filtered photo gallery with a keyboard- and touch-navigable lightbox.

**Restaurant Analytics Suite** *Jan 2025 – Mar 2025*

*Cognifyz Technologies — Internship Project*

- End-to-end restaurant data project covering rating prediction, recommendation, cuisine classification, and location-based geographic analysis, presented as a unified deliverable

**Computer Vision Mini-Projects** *2024 – 2025*

*Personal & Coursework*

- Built MNIST digit recognition with CNNs, image classification with HOG + SVM, and basic object/face detection demos using OpenCV; explored data augmentation strategies and model evaluation

## CERTIFICATIONS & MEMBERSHIPS

- Machine Learning Specialization — Stanford University (Coursera)
- Deep Learning for Computer Vision — Online Certification
- IEEE Student Member, University of Malakand Chapter

## REFERENCES

Dr. Khalil Ullah
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